

Cryptanalysis of a multivariate CCZ scheme

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Abstract. We consider the multivariate scheme **Pesto**, which was introduced by Calderini, Caminata, and Villa. In this scheme, the public polynomials are obtained by applying a CCZ transformation to a set of quadratic secret polynomials. As a consequence, the public key consists of polynomials of degree 4. In this work, we show that the public degree 4 polynomial system can be efficiently reduced to a system of quadratic polynomials. This seems to suggest that the CCZ transformation may not offer a significant increase in security, contrary to what was initially believed.

1 Introduction

Multivariate cryptography is one of the main candidates for providing quantum resistant public key primitives. The security of multivariate cryptography is based on the Polynomial System Solving Problem (PoSSo Problem), that is, solving a random system of multivariate polynomials of degree at least 2. This problem is known to be NP-complete and believed to be hard in the average case, already for quadratic polynomials over the field $\mathbb{F}_2$.

In the usual setup of multivariate schemes, there is a central (secret) map F consisting of m polynomials in n variables. These polynomials (usually quadratic) are not random, but rather possess a trapdoor that allows the legitimate user to easily find a solution to the system. The map F is then composed with two randomly chosen affine linear bijections, A_1 and A_2 , respectively over $\mathbb{F}_q^m$ and $\mathbb{F}_q^n$, to obtain the public map $G_{\text{pub}} = A_1 \circ F \circ A_2$. The public map consists of polynomials that resemble random polynomials and is therefore assumed to be hard to invert. The secret and public keys, F and G_{pub} , are called *affine equivalent*.

Despite the use of affine random bijections, the maps F and G_{pub} often share algebraic properties, that have been exploited in the past to attack several proposed cryptographic primitives. In an attempt to propose an alternative construction, Calderini, Caminata, and Villa [2] suggested applying more general transformations to the central map, in order to obtain a public-key function G_{pub} that does not directly inherit the “simple” structure of the secret function

F . Specifically, they propose using the CCZ transformation, introduced by Carlet, Charpin, and Zinoviev [6] in the context of Boolean functions. One potential advantage of the CCZ transformation is that it usually produces public polynomials of higher degree than the secret polynomials, making them potentially more difficult to solve, while keeping the secret key relatively small. The authors of [2] presented a general instance of the scheme, which can be applied to any central map admitting a t -twist, see [5]. Then, they presented some restrictions on the choice of the quadratic secret map used, summarized in a concrete instance with a specific choice for the central map (see [2, §4.2.3]), which they call *Pesto*, on which we will focus throughout this paper.

In the *Pesto* scheme, the central map consists of quadratic equations in n variables that, once composed with the CCZ transformation, contains equations of degree up to 4. Thus, the public key G_{pub} is of degree 4, while the secret key F is of degree 2. In this paper, we show that the equations of the public key can be efficiently reduced to equations of degree 2. This suggests that there is no clear advantage in terms of security in composing with the CCZ transformation.

In this paper, we discuss two techniques for carrying on this reduction: Gröbner bases algorithms and Higher Order Linearization Equations. The second method seems to be asymptotically faster, with a complexity of $\mathcal{O}(n^7)$, when $m \leq n$. Both methods take advantage of the structure of the central map for the reduction. Once G_{pub} is reduced to a quadratic polynomial system, it can be solved by using general quadratic solving methods or any OV solving methods, if it has the structure of an Oil and Vinegar (OV) scheme.

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2 Preliminaries and notation

Let $\mathbb{F}_q$ denote the finite field with q elements and let $\mathbb{F}_q[z_1, \dots, z_n]$ be the polynomial ring in n variables over $\mathbb{F}_q$. Given a system $F = \{f_1, \dots, f_m\}$ of m equations in $\mathbb{F}_q[z_1, \dots, z_n]$, we denote by F^{TOP} the system $\{f_1^{\text{TOP}}, \dots, f_m^{\text{TOP}}\}$, where f_i^{TOP} is the homogeneous component of maximum degree of f_i . Moreover, we denote by $\langle F \rangle$ the vector space generated by F , by (F) the ideal generated by F , and by $(F)_d$ the vector space of homogeneous elements of degree d of (F) .

2.1 The Pesto scheme

In this section we briefly describe the multivariate scheme **Pesto**, proposed by Calderini, Caminata and Villa and provide an overview of the key concepts required. For more detailed information, we refer to the original article [2].

The central map of **Pesto** $F: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ is a quadratic polynomial map of the form $F(x, y) = (T(x, y), U(x, y))$, where both T and U have degree at most 2.

Definition 1 (Pesto). Fix positive integers n, m, t , and s with $t \leq \min\{n, m\}$ and $0 < s \leq n - t$, then consider the maps:

1. $T: \mathbb{F}_q^t \times \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^t$ defined as $T(x, y) = x + \mathbf{q}(y)$, where $\mathbf{q}: \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^t$ is a random quadratic polynomial map. The inverse of T with respect to x , denoted as $T_y^{-1}(x)$, is then $x - \mathbf{q}(y)$.
2. $U: \mathbb{F}_q^t \times \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^{m-t}$ a system of $m - t$ random Oil and Vinegar (OV) maps [7], with $x_1, \dots, x_t, y_1, \dots, y_s$ vinegar variables and $y_{s+1}, \dots, y_{n-t}$ oil variables. In other words, in **Pesto** the equations of U have the form

$$f^{(i)} = \sum_{j,k \in V} \alpha_{jk}^{(i)} z_j z_k + \sum_{j \in V, k \in O} \beta_{jk}^{(i)} z_j z_k + \sum_{j \in V \cup O} \gamma_j^{(i)} z_j + \delta^{(i)} \quad (1)$$

where $V = \{1, \dots, s + t\}$, $O = \{s + t + 1, \dots, n\}$, $z_i = x_i$ for $1 \leq i \leq t$, $z_i = y_{i-t}$ for $t + 1 \leq i \leq n$. The coefficients $\alpha_{jk}^{(i)}, \beta_{jk}^{(i)}, \gamma_j^{(i)}, \delta^{(i)} \in \mathbb{F}_q$ are chosen uniformly at random.

3. $A_1: \mathbb{F}_q^m \rightarrow \mathbb{F}_q^m$ a random affine linear bijection.
4. $A_2: \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ a random affine linear bijection.

Let $F(x, y) = (x + \mathbf{q}(y), U(x, y))$. The corresponding secret map is the system $G(x, y) = F(T_y^{-1}(x), U(T_y^{-1}(x), y)) = (x - \mathbf{q}(y), U(x - \mathbf{q}(y), y))$. The public key is then given by the map $G_{\text{pub}} = A_1 \circ G \circ A_2$, while the secret key consists of $A_1, A_2, \mathbf{q}, U$.

We assume throughout the paper that each component of $\mathbf{q}$ has degree 2. In fact, if a component of $\mathbf{q}$ is linear, then composing $x - \mathbf{q}(y)$ with A_2 is just a change of coordinates on that component of G . However, the goal of applying a CCZ transformation is to introduce a non-linear transformation in addition to A_2 .

Using this setup, one can design either a digital signature scheme or an encryption scheme, as demonstrated by the authors in [2, §4.1.1 and §4.1.2]. Based on the security analysis performed in [2, §6], the authors recommended that t should be around $n/3$.

2.2 Toy example

We provide a toy example of **Pesto** over $\mathbb{F}_3$, which will be our running example throughout the paper. The example was produced using Magma [1]. The code used to produce it can be found in Appendix A.1.

We let $n = 6$, $m = 5$, $t = 2$, and $s = 1$. Therefore, we have the sets of variables $x = \{x_1, x_2\}$ and $y = \{y_1, y_2, y_3, y_4\}$. In order to define the map $T(x, y): \mathbb{F}_3^2 \times \mathbb{F}_3^4 \rightarrow \mathbb{F}_3^2$, we need to choose a random quadratic map $\mathbf{q}: \mathbb{F}_3^4 \rightarrow \mathbb{F}_3^2$. Let

$$\mathbf{q}(y) = \begin{bmatrix} 2y_1^2 + 2y_1y_2 + 2y_2y_3 + 2y_3y_4 + y_4^2 + y_3 + y_4 + 2 \\ 2y_1y_3 + y_3^2 + y_1y_4 + 2y_3y_4 + 2y_4^2 + 2y_1 + 2y_3 + 2 \end{bmatrix}$$

and let $T(x, y) = x + \mathbf{q}(y)$. The map U is an OV system of 3 equations with vinegar variables x_1, x_2, y_1 and oil variables y_2, y_3, y_4 . We choose

$$U(x, y) = \begin{bmatrix} x_1^2 + x_1x_2 + 2x_1y_1 + 2x_2y_1 + 2y_1^2 + x_1y_2 + x_2y_2 + y_1y_2 + 2x_1y_3 \\ + x_2y_3 + 2y_1y_3 + x_1y_4 + y_1y_4 + 2x_1 + x_2 + y_1 + y_3 + 1 \\ x_1x_2 + 2x_1y_1 + x_2y_1 + y_1^2 + 2x_1y_2 + 2x_2y_3 + 2x_1y_4 \\ + x_1 + x_2 + y_1 + 2y_3 + 2 \\ x_1x_2 + x_2^2 + 2x_1y_1 + y_1^2 + x_1y_2 + y_1y_2 + 2x_1y_3 + y_1y_3 + x_1y_4 \\ + 2x_2y_4 + y_1y_4 + 2x_1 + 2y_1 + 2y_2 + 2y_3 + 2y_4 + 1 \end{bmatrix}.$$

With this choice of T and U , $G(x, y)$ takes the form

$$\begin{bmatrix} y_1^2 + y_1y_2 + y_2y_3 + y_3y_4 + 2y_4^2 + x_1 + 2y_3 + 2y_4 + 1 \\ y_1y_3 + 2y_3^2 + 2y_1y_4 + y_3y_4 + y_4^2 + x_2 + y_1 + y_3 + 1 \\ y_1^4 + 2y_1^3y_2 + y_1^2y_2^2 + y_1^3y_3 + 2y_1y_2^2y_3 + 2y_1^2y_3^2 + y_2^2y_3^2 + 2y_2y_3^3 + 2y_1^3y_4 + 2y_1^2y_2y_4 \\ + 2y_1y_2y_3y_4 + y_1y_3^2y_4 + 2y_3^3y_4 + 2y_1^2y_4^2 + 2y_1y_2y_4^2 + y_1y_3y_4^2 + 2y_2y_3y_4^2 + y_1y_4^3 + y_3y_4^3 \\ + 2x_1y_1^2 + x_2y_1^2 + 2x_1y_1y_2 + x_2y_1y_2 + y_1^2y_2 + y_1y_2^2 + x_1y_1y_3 + 2x_1y_2y_3 + x_2y_2y_3 \\ + 2y_1y_2y_3 + y_2^2y_3 + 2x_1y_3^2 + y_1y_3^2 + 2x_1y_1y_4 + y_1y_2y_4 + x_2y_3y_4 + y_1y_3y_4 + y_2y_3y_4 \\ + 2y_3^2y_4 + 2x_1y_4^2 + 2x_2y_4^2 + x_1^2 + x_1x_2 + 2x_2y_1 + x_1y_2 + x_2y_2 + y_1y_2 + x_1y_3 + y_1y_3 \\ + 2y_2y_3 + 2x_1y_4 + 2x_2y_4 + 2y_1y_4 + 2y_2y_4 + 2y_3y_4 + 2x_1 + 2x_2 + y_1 + 2y_2 + y_3 + 2y_4 \\ y_1^3y_3 + y_1^2y_2y_3 + 2y_1^2y_3^2 + 2y_2y_3^3 + 2y_1^3y_4 + 2y_1^2y_2y_4 + y_1^2y_3y_4 + y_1y_3^2y_4 + y_2y_3^2y_4 \\ + 2y_3^3y_4 + y_1^2y_4^2 + y_1y_2y_4^2 + y_1y_3y_4^2 + y_2y_3y_4^2 + 2y_3^2y_4^2 + y_1y_4^3 + 2y_4^4 + x_2y_1^2 + x_2y_1y_2 \\ + 2y_1^2y_2 + 2y_1y_2^2 + x_1y_1y_3 + 2y_1^2y_3 + x_2y_2y_3 + y_1y_2y_3 + 2y_2^2y_3 + 2x_1y_3^2 + y_2y_3^2 + 2y_3^3 \\ + 2x_1y_1y_4 + y_1^2y_4 + 2y_1y_2y_4 + x_1y_3y_4 + x_2y_3y_4 + 2y_1y_3y_4 + y_2y_3y_4 + x_1y_4^2 + 2x_2y_4^2 \\ + 2y_1y_4^2 + y_2y_4^2 + y_3y_4^2 + x_1x_2 + x_2y_1 + y_1^2 + 2x_1y_2 + 2y_1y_2 + x_1y_3 + x_2y_3 + 2y_1y_3 \\ + 2y_3^2 + 2x_1y_4 + 2x_2y_4 + y_1y_4 + y_2y_4 + y_3y_4 + y_4^2 + 2x_1 + 2x_2 + 2y_2 + y_3 + 2 \\ y_1^3y_3 + y_1^2y_2y_3 + y_1y_3^3 + 2y_2y_3^3 + y_3^4 + 2y_1^3y_4 + 2y_1^2y_2y_4 + 2y_1^2y_3y_4 + 2y_1y_3^2y_4 \\ + y_2y_3^2y_4 + 2y_1^2y_4^2 + y_1y_2y_4^2 + y_1y_3y_4^2 + y_2y_3y_4^2 + y_3^2y_4^2 + 2y_1y_4^3 + 2y_3y_4^3 + x_2y_1^2 \\ + x_2y_1y_2 + y_1^2y_2 + y_1y_2^2 + x_1y_1y_3 + 2x_2y_1y_3 + 2y_1^2y_3 + x_2y_2y_3 + y_2^2y_3 + 2x_1y_3^2 + x_2y_3^2 \\ + 2y_1y_3^2 + 2y_3^3 + 2x_1y_1y_4 + x_2y_1y_4 + 2y_1^2y_4 + y_1y_2y_4 + x_1y_3y_4 + 2y_1y_3y_4 + 2y_2y_3y_4 \\ + x_1y_4^2 + x_2y_4^2 + y_1y_4^2 + 2y_2y_4^2 + x_1x_2 + x_2^2 + 2x_2y_1 + 2y_1^2 + x_1y_2 + y_1y_2 + x_2y_3 \\ + 2y_2y_3 + y_3^2 + x_1y_4 + x_2y_4 + 2y_2y_4 + y_3y_4 + 2y_4^2 + y_1 + y_3 + 2y_4 + 2 \end{bmatrix}.$$

Finally, we choose the following affine linear bijections of $\mathbb{F}_3^5$ and $\mathbb{F}_3^6$

$$A_1(x, y) = \begin{bmatrix} 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 2 & 2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}, \quad A_2(x, y) = \begin{bmatrix} 0 & 2 & 2 & 0 & 2 & 0 \\ 0 & 2 & 1 & 2 & 2 & 1 \\ 0 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \\ 0 \\ 2 \end{bmatrix}.$$

The functions $\mathbf{q}$, U , A_1 , A_2 were randomly generated in Magma [1]. The public map $G_{\text{pub}} = A_1 \circ G \circ A_2$ consists of 5 dense polynomials of degree 4. Given their sizes, we opt not to report them here. To conclude, we choose a random secret input $\text{ipt} = (2, 0, 1, 2, 2, 0) \in \mathbb{F}_3^6$ for the public system G_{pub} and we compute the corresponding output $\text{opt} = (1, 2, 2, 1, 2) \in \mathbb{F}_3^5$.

2.3 Security analysis

In this section, we summarize the security analysis of **Pesto** from [2, §6]. In particular, we focus on some well-known attacks for solving Oil and Vinegar (OV) systems and algebraic attacks using Gröbner Bases, as these attacks will be revisited in the sequel.

Structural attacks The secret map F is constructed using two Oil and Vinegar systems, $T(x, y)$ and $U(x, y)$. This naturally raises the question of whether known attacks on OV systems could also be extended to **Pesto**. After applying the twisting transformation, the first t coordinates of the map G retain the structure of an OV system, $x - \mathbf{q}(y)$, while the remaining $m - t$ coordinates are usually polynomials of degree 4. Therefore, if it is possible to isolate the components of G_{pub} corresponding to $x - \mathbf{q}(y)$, the known attacks to OV systems could be directed specifically at these components. In particular, due to the Kipnis–Shamir attack [10] on balanced Oil and Vinegar signature schemes, the authors of [2] recommend keeping the system $x - \mathbf{q}(y)$ unbalanced. For instance, setting $t \approx n/3$ helps to maintain this imbalance. On the other hand, since the second part of the system G consists of polynomials with degrees up to 4, they argue that standard OV attacks are unlikely to be effective against the entire map G_{pub} . Additionally, they present a linearization attack that can be performed when $s = 0$. Therefore, they recommend choosing $s > 0$.

Algebraic attacks using Gröbner bases We consider the scenario where an attacker aims to forge a signature by finding a preimage $v \in \mathbb{F}_q^n$ of a given element $w \in \mathbb{F}_q^m$. This can be done by computing a Gröbner basis for the polynomial system G_{pub} , see e.g. [3] for details on how this can be done. The asymptotic complexity of these methods is

$$O\left(\binom{n + \text{sd}(G_{\text{pub}})}{n}\right)^\omega,$$

where n is the number of variables, $\text{sd}(G_{\text{pub}})$ is the solving degree of the system G_{pub} , and $2 < \omega < 3$ is the matrix-multiplication exponent.

Informally, the solving degree is the maximum degree of the polynomials produced during the Gröbner basis computation, see [4] for a precise definition and more details. In order to estimate the solving degree of the system G_{pub} associated with **Pesto**, the authors of [2] conducted computational experiments in Magma [1] for different choices of the parameters n, m, t, s, q . A summary of the results may be found in [2, §6.6]. It appears that the solving degree increases rapidly with respect to the input parameters, making this attack quickly infeasible.

3 Reducing the public key to a quadratic system

In this section, we show how to reduce the public key G_{pub} associated to a system of the form $F(x, y) = (T(x, y), U(x, y))$ to a quadratic system, under the assumption that U, T , and T_y^{-1} are quadratic. This applies in particular to **Pesto**, see Definition 1. We present two methods for carrying on the reduction: one using a linear algebra based algorithm for Gröbner bases in §3.1, and the other via Higher Order Linearization Equations (HOLE) in §3.2.

3.1 Method 1: Gröbner Bases Algorithms

Throughout this subsection, we assume that T has the form $T = x + \mathbf{q}(y)$ as in Definition 1. We aim to demonstrate that by running a linear algebra based algorithm on the Macaulay matrix of degree 4 associated to G_{pub} , we can reduce G_{pub} to a system of quadratic equations. To establish this result, we will use the framework of constructible polynomials. We begin by recalling the relevant definitions and some useful properties from [9].

Definition 2. Let $\mathcal{F} \subseteq \mathbb{F}_q[z_1, \dots, z_n]$ be a system of polynomials and let $i \in \mathbb{N}$. With $\mathbb{F}_q[z_1, \dots, z_n]_{\leq i}$ we denote the set of polynomials of degree up to i . The space $V_{\mathcal{F}, i}$ of constructible polynomials is the smallest $\mathbb{F}_q$ -vector space such that

1. $\mathcal{F} \cap \mathbb{F}_q[z_1, \dots, z_n]_{\leq i} \subseteq V_{\mathcal{F}, i}$;
2. if $g \in V_{\mathcal{F}, i}$ and $f \in \mathbb{F}_q[z_1, \dots, z_n]$ with $\deg(fg) \leq i$ then $fg \in V_{\mathcal{F}, i}$.

Given two polynomials f, g we write $f \equiv_{\mathcal{F}, i} g$ if $f - g \in V_{\mathcal{F}, i}$.

In the following remark, we collect some properties from [9, Proposition 2.3], which we will use in the sequel.

Remark 1. Let $\mathcal{F}, \mathcal{G} \subseteq \mathbb{F}_q[z_1, \dots, z_n]$ be systems of polynomials, let $i \in \mathbb{N}$, and let $A : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ be an affine linear bijection. Then, the following properties hold:

1. If $\mathcal{F} \subseteq \mathcal{G}$ then $V_{\mathcal{F}, i} \subseteq V_{\mathcal{G}, i}$;
2. $V_{\mathcal{F}, i} \circ A = V_{\mathcal{F} \circ A, i}$.

Lemma 1. *Let $U \subseteq \mathbb{F}_q[x_1, \dots, x_t, y_1, \dots, y_{n-t}]$ and $\mathbf{q} = \{\mathbf{q}_1, \dots, \mathbf{q}_t\} \subseteq \mathbb{F}_q[y_1, \dots, y_{n-t}]$ be systems of quadratic equations and let u be an equation of the system $U(x - \mathbf{q}(y), y)$. Then $u \equiv_{x-\mathbf{q}(y),4} v$ with $\deg v \leq 2$.*

Proof. Assuming that the i -th equation of $U(x, y)$ has the form

$$\sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} x_j x_k + \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k} x_j y_k + \sum_{j=1}^{n-t} \sum_{k=j}^{n-t} \gamma_{j,k} y_j y_k + \sum_{j=1}^t \delta_j x_j + \sum_{j=1}^{n-t} \epsilon_j y_j + \zeta,$$

the i -th equation of $U(x - \mathbf{q}(y), y)$ is of the form

$$\begin{aligned} u = & \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} x_j x_k - \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} x_j \mathbf{q}_k - \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_j x_k + \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_j \mathbf{q}_k \\ & + \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k} x_j y_k - \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k} \mathbf{q}_j y_k + \sum_{j=1}^{n-t} \sum_{k=j}^{n-t} \gamma_{j,k} y_j y_k + \sum_{j=1}^t \delta_j x_j \\ & - \sum_{j=1}^t \delta_j \mathbf{q}_j + \sum_{j=1}^{n-t} \epsilon_j y_j + \zeta, \end{aligned}$$

where we removed the dependency on i from the coefficients and on the input variables y from the $\mathbf{q}_j$'s to ease the notation. We denote by u^r and $\mathbf{q}_j^r$ the homogeneous component of degree r of u and $\mathbf{q}_j$, respectively. Therefore, we have

$$\begin{aligned} u^4 &= \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_j^2 \mathbf{q}_k^2 \\ u^3 &= \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_j^2 \mathbf{q}_k^1 + \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_j^1 \mathbf{q}_k^2 - \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} x_j \mathbf{q}_k^2 \\ &\quad - \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_j^2 x_k - \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k} \mathbf{q}_j^2 y_k = \sum_{j=1}^t \mathbf{q}_j^2 h_j^1(x, y), \end{aligned}$$

where $h_j^1(x, y)$ is a suitable homogeneous linear polynomial, for each $j = 1, \dots, t$. Let $p_j = \sum_{k=j}^t \alpha_{j,k} \mathbf{q}_k^2 + h_j^1(x, y)$. Then $u^4 + u^3 = \sum_{j=1}^t p_j \mathbf{q}_j^2$ and

$$\begin{aligned} p_j &= \sum_{k=j}^t \alpha_{j,k} (\mathbf{q}_k^2 + \mathbf{q}_k^1 - x_k) + \sum_{k=1}^j \alpha_{k,j} (\mathbf{q}_k^1 - x_k) - \sum_{k=1}^{n-t} \beta_{j,k} y_k \\ &= \sum_{k=j}^t \alpha_{j,k} (\mathbf{q}_k(y) - x_k - \mathbf{q}_k^0) + \sum_{k=1}^j \alpha_{k,j} (\mathbf{q}_k^1 - x_k) - \sum_{k=1}^{n-t} \beta_{j,k} y_k \equiv_{x-\mathbf{q}(y),2} \ell_j, \end{aligned}$$

where $\deg(\ell_j) \leq 1$. Hence

$$\begin{aligned} u &\equiv_{x-\mathbf{q}(y),4} u + \sum_{j=1}^t p_j(x_j - \mathbf{q}_j) = u^2 + u^1 + u^0 + \sum_{j=1}^t p_j(x_j - \mathbf{q}_j^1 - \mathbf{q}_j^0) \\ &\equiv_{x-\mathbf{q}(y),4} u^2 + u^1 + u^0 + \sum_{j=1}^t \ell_j(x_j - \mathbf{q}_j^1 - \mathbf{q}_j^0), \end{aligned}$$

an element of degree up to 2.

We are now ready to present the reduction of the public system G_{pub} of *Pesto* to a quadratic system, which we formalize in the next theorem.

Theorem 1. *Consider $T : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^t$ such that $T(x, y) = x + \mathbf{q}(y)$, where $\mathbf{q}(y)$ is quadratic. Consider $U : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^{m-t}$ a system of quadratic equations. For $G = (T_y^{-1}(x), U(T_y^{-1}(x), y))$, set $G_{\text{pub}} = A_1 \circ G \circ A_2$, with A_1, A_2 linear affine bijections of $\mathbb{F}_q^m$ and $\mathbb{F}_q^n$ respectively. Starting from the equations of G_{pub} and $c \in \mathbb{F}_q^m$, we can produce:*

- 1) *a linear space of quadratic equations which contains the equations of $(x - \mathbf{q}(y)) \circ A_2$ with complexity $\mathcal{O}(m^2 n^4)$,*
- 2) *a quadratic system with the same solutions as $G_{\text{pub}} = c$ with complexity $\mathcal{O}(n^{16})$,*

Proof. 1) Let W be the subspace of $\langle G_{\text{pub}} \rangle$ consisting of the equations of degree at most 2. Since $\langle G_{\text{pub}} \rangle = \langle G \circ A_2 \rangle$, W contains the t linearly independent quadratic equations of $(x - \mathbf{q}(y)) \circ A_2$.

Build a matrix M associated to the system G_{pub} with respect to a degree-compatible term order as follows. Its columns correspond to the monomials of degree up to 4 in decreasing order with respect to the chosen order. In particular, each column corresponds to a monomial of degree smaller than or equal degree to the monomial corresponding to the column on its left. Each row corresponds to one of the equations of G_{pub} and the entry in the row corresponding to g and in the column corresponding to the monomial η is the coefficient of η in g . Therefore, there is a natural identification between the rowspace of M and $\langle G_{\text{pub}} \rangle$.

Put M in reduced row-echelon form. Notice that now M has the property that each row corresponds to a polynomial of degree smaller than or equal to the polynomial corresponding to the row above. Since M is in reduced row-echelon form, W is the vector space generated by the rows which correspond to polynomials of degree at most 2, showing that the last $\dim(W)$ rows of M are a basis of W . The complexity of computing it is $\mathcal{O}(m^2 n^4)$, i.e., the complexity of Gaussian elimination in M .

2) For ease of notation and up to incorporating the constant in the equations of G_{pub} , we may assume without loss of generality that $c = 0$. To produce a system of quadratic equations with the same solutions as G_{pub} , we consider the Macaulay matrix $\mathcal{M}$ of degree 4 associated to the system G_{pub} with respect to

a degree-compatible term order. Notice that the matrix M constructed in 1) is a submatrix of $\mathcal{M}$. Recall that $\mathcal{M}$ is built as follows. Its columns correspond to the monomials of degree up to 4 ordered in decreasing order with respect to the chosen order and the rows correspond to all the equations μg , where g is one the equations of G_{pub} and μ is a monomial such that $\deg(\mu g) \leq 4$. The entry in the row corresponding to μg and in the column corresponding to the monomial η is the coefficient of η in μg . Throughout the proof, we abuse terminology and we identify the rows of the matrix with the corresponding polynomials.

We run a Gröbner basis-type computation on $\mathcal{M}$ by computing its reduced row-echelon form, together with the mutant trick. More precisely, we perform Gaussian elimination on $\mathcal{M}$ and, whenever a new polynomial f with $\deg(f) < 4$ appears, then we append to the matrix new rows corresponding to μf , for every monomial μ such that $\deg(\mu f) \leq 4$. Then we continue with Gaussian elimination on $\mathcal{M}$. This procedure eventually terminates, since the number of rows in the reduced row-echelon form of the resulting matrix is bounded from above by the number of monomials of degree up to 4. It follows from the proof of [13, Proposition 3.13] that the complexity of this computation is $\mathcal{O}(n^{16})$, since $\mathcal{M}$ has $\mathcal{O}(n^4)$ columns.

We claim that the procedure described above produces a system, which generates the same ideal as G_{pub} . This follows from the fact that computing the reduced row-echelon form of a matrix does not change the vector space, hence the ideal, generated by its rows. Moreover, appending to a matrix rows which are multiples of other rows does not change the ideal generated by its rows. In particular, the algorithm described produces a system which has the same solutions as G_{pub} . Since the term order is degree-compatible, by [8, Theorem 1] the rowspace of the Macaulay matrix coincides with the vector space $V_{G_{\text{pub}},4}$ of constructible polynomials of degree 4 of G_{pub} .

We conclude by proving that the system produced is quadratic. Consider any row g of $\mathcal{M}$ which has degree 3 or 4 after the first round of Gaussian elimination. We claim that $g \equiv_{G_{\text{pub}},4} g'$, for some g' with $\deg(g') \leq 2$. In 1) we showed that, after the first round of Gaussian elimination, the matrix contains rows which generate all the elements of $\langle (x - q(y)) \circ A_2 \rangle$. Thus, by Remark 1 we have $V_{(x-q(y)) \circ A_2,4} \subseteq V_{G_{\text{pub}},4}$, so it suffices to show that $g \equiv_{(x-q(y)) \circ A_2,4} g'$. Equivalently, it suffices to show that $g \circ A_2^{-1} \equiv_{(x-q(y)),4} g' \circ A_2^{-1}$, since an affine linear bijection preserves the degree. Since $g \circ A_2^{-1}$ is a linear combination of the equations of U , we conclude by Lemma 1.

Example 1. Running the algorithm from Theorem 1, 2) on the Toy Example from §2.2 yields the following reduced quadratic public system.

$$G_{\text{red}} = \begin{bmatrix} 2x_1x_2 + 2x_1y_1 + x_2y_1 + 2x_2y_2 + 2x_2y_3 + y_1y_3 + 2y_2y_3 + 2y_3^2 + 2x_1y_4 \\ + x_2y_4 + 2y_2y_4 + y_3y_4 + y_4^2 + 2x_1 + x_2 + y_1 + y_2 + 2y_3 \\ \\ 2x_1^2 + 2x_1x_2 + x_2^2 + 2x_2y_1 + y_1^2 + 2x_1y_2 + x_2y_2 + y_1y_2 + y_2^2 + 2x_1y_3 \\ + 2x_1y_4 + 2x_2y_4 + 2y_2y_4 + 2y_3y_4 + x_1 + y_1 + y_2 + 2y_3 + y_4 + 2 \\ \\ 2x_2^2 + x_2y_1 + x_1y_2 + y_1y_2 + 2y_2^2 + y_1y_3 + 2y_2y_3 + y_3^2 + x_1y_4 + x_2y_4 \\ + 2y_1y_4 + 2y_2y_4 + 2y_3y_4 + 2y_4^2 + x_1 + 2y_1 + 2y_2 + 2y_3 + y_4 + 2 \\ \\ 2x_2y_1 + y_1^2 + 2x_1y_2 + 2x_2y_2 + 2y_1y_2 + x_2y_3 + 2y_3^2 + 2x_1y_4 \\ + 2x_2y_4 + y_1y_4 + y_2y_4 + y_3y_4 + y_4^2 + y_1 + y_3 + 2y_4 \\ \\ x_2^2 + 2x_2y_1 + 2y_1^2 + x_1y_2 + 2x_2y_2 + y_2^2 + y_1y_3 + 2y_2y_3 + y_3^2 + x_1y_4 \\ + x_2y_4 + 2y_1y_4 + 2y_2y_4 + 2y_3y_4 + 2y_4^2 + x_2 + y_1 + y_2 + y_3 + 1 \end{bmatrix}.$$

3.2 Method 2: A Higher Order Linearization Equation (HOLE) attack

In this section, we recall the linearization attack and we derive a similar method to reduce the public system to a system of quadratic equations.

Higher order linearization equations (HOLEs)

Definition 3. Consider $\mathcal{P} = (p_1, \dots, p_m) : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ the public key of a multivariate public key primitive, with $p_i \in \mathbb{F}_q[x_1, \dots, x_n]$ for $1 \leq i \leq m$.

A (valid) input/output pair is a pair $(\bar{z}, \bar{w}) \in \mathbb{F}_q^n \times \mathbb{F}_q^m$ such that $\mathcal{P}(\bar{z}) = \bar{w}$.

A linearization equation is a polynomial equation in $\mathbb{F}_q[z_1, \dots, z_n, w_1, \dots, w_m]$ of the form

$$\mathcal{R}(z, w) = \sum_{i=1}^m \sum_{j=1}^n \alpha_{ij} w_i z_j + \sum_{i=1}^m \beta_i w_i + \sum_{j=1}^n \gamma_j z_j + \delta, \quad (2)$$

such that $\mathcal{R}(\bar{z}, \bar{w}) = 0$ for any valid input/output pair $(\bar{z}, \bar{w})$.

A higher order linearization equation (HOLE) is a polynomial equation of the form

$$\mathcal{R}(z, w) = \sum_{j=1}^n g_j(w_1, \dots, w_m) z_j + g(w_1, \dots, w_m), \quad (3)$$

with $g, g_j \in \mathbb{F}_q[w_1, \dots, w_m]$ for $1 \leq j \leq n$, such that $\mathcal{R}(\bar{z}, \bar{w}) = 0$ for any valid input/output pair $(\bar{z}, \bar{w})$. The degree of the HOLE $\mathcal{R}$ is given by

$$d = \max\{\deg(g_1), \dots, \deg(g_n), \deg(g)\}.$$

Notice that, when evaluating (2) and (3) at a ciphertext (output) $(\bar{w}_1, \dots, \bar{w}_m)$, one obtains linear equations in the plaintext (input) variables

$z_1, \dots, z_n$, that is, equations of the form $\mathcal{R}(z, \bar{w}) = 0$.

The existence of linearization equations for a public key $\mathcal{P}$ allows to perform the so-called *linearization equation attack*. This attack was used by Patarin in [12] to break the Matsumoto-Imai cryptosystem [11], one of the first multivariate public-key cryptosystem proposed.

The idea of the attack is the following. Suppose that we have a public key $\mathcal{P}$ and one knows that it satisfies some linearization equations as in (2). Then by computing enough input/output pairs and substituting them into the linearization equation, one gets a system of linear equations in the coefficients $\alpha_{ij}, \beta_i, \gamma_j$ and δ , which can be solved by Gaussian elimination. In case of HOLEs as in (3), the number of needed input/output pairs increases exponentially with the degree d . For example, for $d = 1$ one needs approximately $(m+1) \cdot (n+1)$, while for $d = 2$ one needs approximately $(n+1) \cdot \frac{(m+1)(m+2)}{2}$ input/output pairs.

After performing Gaussian elimination, one gets a system of equations in the input/output variables $z_1, \dots, z_n, w_1, \dots, w_m$. Now, assume that one wishes to decrypt a ciphertext $\bar{w} = (\bar{w}_1, \dots, \bar{w}_m)$. By substituting these values into the previously computed equations, one obtains linear equations in the plaintext variables $z_1, \dots, z_n$. Sometimes, one gets enough linear equations to recover the plaintext. Else, the linear equations can be used to compute partial information on the input, which in turn may be used to speed up other attacks against the cryptosystem.

Recovering more quadratic equations with the HOLE-technique We briefly recall the setup from Section 2. For $T : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^t$ such that $T(x, y) = T_y(x)$ is invertible for any fixed y , and $U : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^{m-t}$ quadratic, we let

$$\begin{aligned} F(x, y) &= (T_y(x), U(x, y)), \\ G(x, y) &= (T_y^{-1}(x), U(T_y^{-1}(x), y)). \end{aligned}$$

Notice that for $G(x, y) = (w_T, w_U) \in \mathbb{F}_q^t \times \mathbb{F}_q^{m-t}$, it holds $w_U = U(w_T, y)$.

Consider an arbitrary quadratic system $U(x, y)$ of $m - t$ polynomials of the form

$$\sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k}^{(i)} x_j x_k + \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k}^{(i)} x_j y_k + \sum_{j=1}^{n-t} \sum_{k=j}^{n-t} \gamma_{j,k}^{(i)} y_j y_k + \sum_{j=1}^t \delta_j^{(i)} x_j + \sum_{j=1}^{n-t} \epsilon_j^{(i)} y_j + \zeta^{(i)},$$

for $1 \leq i \leq m - t$. We introduce the input and output variables $x = (z_1, \dots, z_t)$, $y = (z_{t+1}, \dots, z_n)$, $w_T = (w_1, \dots, w_t)$ and $w_U = (w_{t+1}, \dots, w_m)$. The relation $w_U = U(w_T, y)$ translates into the quadratic relations

$$\begin{aligned} w_{t+i} &= \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k}^{(i)} w_j w_k + \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k}^{(i)} w_j z_{k+t} + \sum_{j=1}^{n-t} \sum_{k=j}^{n-t} \gamma_{j,k}^{(i)} z_{j+t} z_{k+t} \\ &\quad + \sum_{j=1}^t \delta_j^{(i)} w_j + \sum_{j=1}^{n-t} \epsilon_j^{(i)} z_{j+t} + \zeta^{(i)}, \end{aligned}$$

for $1 \leq i \leq m - t$. Let

$$\begin{aligned} \mathcal{R}_G^{(i)}(z, w) = & \sum_{j=1}^t \sum_{k=j}^t \alpha_{j,k}^{(i)} w_j w_k + \sum_{j=1}^t \sum_{k=1}^{n-t} \beta_{j,k}^{(i)} w_j z_{k+t} + \sum_{j=1}^{n-t} \sum_{k=j}^{n-t} \gamma_{j,k}^{(i)} z_{j+t} z_{k+t} \\ & + \sum_{j=1}^t \delta_j^{(i)} w_j + \sum_{j=1}^{n-t} \epsilon_j^{(i)} z_{j+t} + \zeta^{(i)} - w_{i+t}, \end{aligned} \quad (4)$$

for $1 \leq i \leq m - t$. The equations (4) satisfy $\mathcal{R}_G^{(i)}(\bar{z}, \bar{w}) = 0$ for $1 \leq i \leq m - t$ and any input/output pair $(\bar{z}, \bar{w})$ such that $G(\bar{z}) = \bar{w}$.

Remark 2. Notice that the $m - t$ relations $\mathcal{R}_G^{(1)}(z, w), \dots, \mathcal{R}_G^{(m-t)}(z, w)$ are linearly independent, since $\mathcal{R}_G^{(i)}(z, w)$ is the only one among them that involves the variable w_{i+t} .

Since G_{pub} is obtained from G by composing with affine linear bijections, we have $m - t$ linearly independent quadratic relations also for the public key. These relations take the form

$$\begin{aligned} \mathcal{R}_{\text{pub}}^{(i)}(z, w) = & \sum_{j=1}^m \sum_{k=j}^m \bar{\alpha}_{j,k}^{(i)} w_j w_k + \sum_{j=1}^m \sum_{k=1}^n \bar{\beta}_{j,k}^{(i)} w_j z_k + \sum_{j=1}^n \sum_{k=j}^n \bar{\gamma}_{j,k}^{(i)} z_j z_k \\ & + \sum_{j=1}^m \bar{\delta}_j^{(i)} w_j + \sum_{j=1}^n \bar{\epsilon}_j^{(i)} z_j + \bar{\zeta}, \end{aligned} \quad (5)$$

for $i = 1, \dots, m - t$.

In order to recover the coefficients of $\mathcal{R}_{\text{pub}}^{(i)}(w, z)$, we need to compute around $\binom{n+m+2}{2}$ input/output pairs and then perform Gaussian elimination on a matrix of approximately the same size.

Once the relations $\mathcal{R}_{\text{pub}}^{(i)}(z, w)$ are found, we evaluate all relations at the target value $\bar{w}$, obtaining the relations in the input variables

$$\mathcal{R}_{\text{pub}}^{(1)}(z, \bar{w}), \dots, \mathcal{R}_{\text{pub}}^{(m-t)}(z, \bar{w}).$$

Remark 3. We stress that the effectiveness of using HOLE-techniques to recover more quadratic equations does not depend on a specific choice of U and T . We only assume that U is a system of quadratic equations. No assumption on T is needed, beyond being invertible. In the sequel, we analyze in more detail the case when $T_y^{-1}(x)$ has degree two.

If T is such that $T_y^{-1}(x)$ is quadratic, then the relation $w_T = T_y^{-1}(x)$ yields t linearly independent quadratic relations between the input and the output of G , implying the existence of other t linearly independent quadratic relations between the input and the output of G_{pub} . The equations are linearly independent also from the ones that we previously found, since they do not involve $w_{t+1}, \dots, w_m$ and the i -th among these additional equations is the only one

involving w_i for $1 \leq i \leq m$. Therefore, when performing the Gaussian elimination on the matrix used to recover the relations (5), we obtain a set of solutions of dimension at least m . Notice that this is the case for **Pesto**, where $T(x, y) = x + \mathbf{q}(y)$ and $T_y^{-1}(x) = x - \mathbf{q}(y)$, where $\mathbf{q} : \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^t$ is quadratic.

We now explain how to isolate within this set of relations those that correspond to the first t equations of G , i.e., to $w_T = T_y^{-1}(x)$. Since $T_y^{-1}(x)$ is quadratic, $w_T = T_y^{-1}(x)$ corresponds to t relations of the form

$$\sum_{j,k=1}^n \bar{\gamma}_{j,k} z_j z_k + \sum_{j=1}^m \bar{\delta}_j w_j + \sum_{j=1}^n \bar{\epsilon}_j z_j + \bar{\zeta}, \quad (6)$$

where $z_1, \dots, z_n$ are the input variables and $w_1, \dots, w_m$ the output variables. Since G and G_{pub} are related by two linear affine bijections, one in the input and one in the output variables, the input/output pairs of G_{pub} also satisfy t relations of the form (6). Therefore, the linear space of relations (5) for G_{pub} contains at least t relations of the form (6):

$$\mathcal{R}_{pub}^{(i)}(z, w) = \sum_{j=1}^n \sum_{k=j}^n \bar{\gamma}_{j,k}^{(i)} z_j z_k + \sum_{j=1}^m \bar{\delta}_j^{(i)} w_j + \sum_{j=1}^n \bar{\epsilon}_j^{(i)} z_j + \bar{\zeta}.$$

In order to find them, one can take a basis of the space of the quadratic relations of G_{pub} and write its elements as rows of a matrix, whose columns correspond to the monomials of degree up to 2 in z, w ordered in degree-lexicographic or degree-reverse-lexicographic order with $w_1 > \dots > w_m > z_1 > \dots > z_n$. Since the row space of the matrix contains a vector space of dimension $d \geq t$ of equations which do not involve any of the monomials $w_j w_k$ and $w_j z_k$, the last d rows of the reduced row-echelon form of the matrix generate the vector subspace of relations of G_{pub} of the form (5), which contains the desired t relations. Notice that, as in the Gröbner basis approach, if $d > t$, one may proceed with the d equations that were found and is still guaranteed to be able to reduce the system G_{pub} to a quadratic system.

Theorem 2. Consider $T : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^t$ such that $T(x, y) = T_y(x)$ is invertible for any fixed y and $T_y^{-1}(x)$ is quadratic. Consider $U : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^{m-t}$ a system of quadratic equations. For $G = (T_y^{-1}(x), U(T_y^{-1}(x), y))$, set $G_{pub} = A_1 \circ G \circ A_2$, with A_1, A_2 linear affine bijections of $\mathbb{F}_q^m$ and $\mathbb{F}_q^n$ respectively. Starting from the equations of G_{pub} and $c \in \mathbb{F}_q^m$, we can produce:

- 1) a system of quadratic relations for G_{pub} with complexity $\mathcal{O}(\max\{mn^6, m^3n^4, m^6\})$,
- 2) a quadratic system with the same solutions as $G_{pub} = c$ with additional negligible complexity.

Proof. 1) The system of quadratic relations for G_{pub} , introducing variables also for the output space, can be obtained via Method 2, as already explained. We recall that the method consists of the following steps:

- a) compute around $\binom{n+m+2}{2}$ random evaluation of the public system;
- b) perform Gaussian elimination on the corresponding matrix to recover the coefficients of the quadratic relations.

The complexity of evaluating the public system at a single chosen input is $\mathcal{O}(m\binom{n+4}{4})$, since we have m equations of degree 4 in n variables. This has to be done about $\binom{n+m+2}{2}$ times, in order to produce the system to be solved. The complexity of solving the linear system of approximate size $\binom{n+m+2}{2}$ is $\mathcal{O}(\binom{n+m+2}{2}^3)$. Therefore, the complexity for this method is $\mathcal{O}(m\binom{n+4}{4}\binom{n+m+2}{2} + \binom{n+m+2}{2}^3) = \mathcal{O}(mn^6 + m^3n^4 + m^6)$.

- 2) To produce the desired quadratic system, we evaluate the output variables at c . Hence we evaluate $\binom{m+2}{2} + mn$ terms. The complexity of this final operation is $\mathcal{O}(m\binom{m+2}{2} + m^2 \cdot n) = \mathcal{O}(m^3 + m^2n)$.

Example 2. Performing the attack explained in Theorem 2 on the Toy Example from §2.2 yields the reduced quadratic public system

$$G_{\text{red}} = \begin{bmatrix} y_1^2 + x_1y_2 + x_2y_2 + 2y_1y_2 + y_1y_3 + 2y_2y_3 + y_3^2 + x_1y_4 + x_2y_4 \\ + 2y_1y_4 + 2y_2y_4 + 2y_3y_4 + 2y_4^2 + 2x_1 + 2x_2 + 2y_4 \\ 2x_2y_1 + x_1y_2 + x_2y_2 + x_2y_3 + 2y_1y_3 + y_2y_3 + y_3^2 + x_1y_4 + x_2y_4 \\ + 2y_1y_4 + 2y_2y_4 + 2y_3y_4 + 2y_4^2 + x_1 + x_2 + y_1 + y_3 \\ x_2^2 + 2x_2y_1 + 2y_1^2 + x_1y_2 + 2x_2y_2 + y_2^2 + y_1y_3 + 2y_2y_3 + y_3^2 + x_1y_4 \\ + x_2y_4 + 2y_1y_4 + 2y_2y_4 + 2y_3y_4 + 2y_4^2 + x_2 + y_1 + y_2 + y_3 + y_4 + 1 \\ x_1^2 + 2x_2^2 + 2x_1y_1 + 2x_2y_1 + 2y_1^2 + x_1y_2 + x_2y_2 + 2y_1y_2 + 2y_2^2 + x_1y_3 \\ + 2x_2y_3 + y_1y_3 + 2y_2y_3 + 2y_3^2 + 2x_2y_4 + 2y_3y_4 + y_4^2 + x_1 + x_2 + 2y_4 + 1 \\ x_1x_2 + x_1y_1 + 2x_2y_1 + x_2y_2 + x_2y_3 + 2y_1y_3 + y_2y_3 + y_3^2 + x_1y_4 \\ + 2x_2y_4 + y_2y_4 + 2y_3y_4 + 2y_4^2 + x_1 + 2x_2 + 2y_1 + 2y_2 + y_3 \end{bmatrix}.$$

In the following remark, we point out that for the **Pesto** scheme the quadratic system obtained as output of Theorem 2 2) is affine equivalent to an (unbalanced) Oil and Vinegar system.

Remark 4. We keep the same notations as in Theorem 2 and we denote by $z = (x, y)$ the input variables, by $w = (w_T, w_U)$ the output variables, and by $S(x, y) = T_y^{-1}(x)$. By applying the HOLE technique, the relations we get for G are a linear combination of $w_T - S(x, y) = 0$ and $w_U - U(w_T, y) = 0$. Now, since $G_{\text{pub}} = A_1 \circ G \circ A_2$, setting $w = G_{\text{pub}}(z)$ we obtain $A_1^{-1}(w) = G(A_2(z))$. This corresponds to applying the affine linear bijection A_2 to the input variables and the affine linear bijection A_1^{-1} to the output variables. We split each of these linear maps into two components $A_{1,S}^{-1} : \mathbb{F}_q^t \times \mathbb{F}_q^{m-t} \rightarrow \mathbb{F}_q^t$, $A_{1,U}^{-1} : \mathbb{F}_q^t \times \mathbb{F}_q^{m-t} \rightarrow \mathbb{F}_q^{m-t}$, $A_{2,S} : \mathbb{F}_q^t \times \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^t$, $A_{2,U} : \mathbb{F}_q^t \times \mathbb{F}_q^{n-t} \rightarrow \mathbb{F}_q^{n-t}$. Thus, for G_{pub} the

relations that we obtain via HOLE technique are a linear combination of

$$\begin{cases} A_{1,S}^{-1}(w) - S \circ A_2(z) = 0 \\ A_{1,U}^{-1}(w) - U(A_{1,S}^{-1}(w), A_{2,U}(z)) = 0 \end{cases}$$

The last step of Method 2 is to evaluate the target output $\bar{w} \in \mathbb{F}_q^m$ in the previous system and solving with respect to z . This amounts to solving a quadratic system of the form

$$\begin{cases} S \circ A_2(z) = A_{1,S}^{-1}(\bar{w}) \\ U(A_{1,S}^{-1}(\bar{w}), A_{2,U}(z)) = A_{1,U}^{-1}(\bar{w}) \end{cases}$$

So, if S and U are (unbalanced) Oil and Vinegar polynomial maps (as in the case of **Pesto**), then the previous system is affine equivalent to an (unbalanced) OV system.

4 Conclusions

Using either Method 1 or Method 2, we can reduce the public system of **Pesto**, originally consisting of quartics, to a quadratic system. More in general, we can reduce to a quadratic system the quartic public system obtained by disguising a quadratic central function using a CCZ transform, as proposed in [2].

Once we obtain a quadratic system via our methods, however, we still must solve it to retrieve the input. If the obtained quadratic system is a disguised OV system, as it is the case for the system obtained by applying Method 2 to **Pesto**, then specific attacks for OV can be used. Otherwise generic attacks, such as those based on Gröbner basis techniques, may be used. One may also try to use tailored attacks which exploit the structure of the central function. We do not however know of a general way of translating an attack on a multivariate public scheme to a scheme where the same central function is disguised using a CCZ transform.

We conclude that the system **Pesto** proposed in [2] is no more secure than the OV system on which it is based. For multivariate schemes with a quadratic central function disguised by means of a CCZ transform, we believe that the increase in degree when producing a public system from a private system via the CCZ transform does not offer an additional layer of security. In fact, we are able to reduce the quartic public system back to a quadratic system. The question of whether disguising a central function via the CCZ transform may make ineffective certain attacks that work when disguising the central function using linear affine transformations remains open.

A Implementation Details

We provide the Magma [1] code used to implement the toy example and to reduce to a quadratic system with both Method 1 and Method 2.

A.1 The Code to Generate the Public System of Pesto

```

// chosen parameters
n:=6; m:=5; t:=2; s:=1; q:=3;

Fq:=GF(q); // finite field
Vn:=VectorSpace(Fq,n); // input space
Vm:=VectorSpace(Fq,m); // output space
P<[x]>:=PolynomialRing(Fq,n,"grevlex");
Q<[z]>:=PolynomialRing(Fq,m,"grevlex");
GLn:=GL(n,Fq); GLm:=GL(m,Fq);

// construct q(y) (t quadratic equations in n-t variables)
qq:=[];
for ind in [1..t] do
    f:=%[x[i1]*x[i2]*Random(Fq) : i1,i2 in [t+1..n] | i1 le i2]+
        %[x[i1]*Random(Fq) : i1 in [t+1..n]]+Random(Fq);
    qq:=Append(qq,f);
end for;

// construct U(x,y) (m-t quadratic equations with t+s variables vinegar and n-t-s oil)
U:=[];
for ind in [1..m-t] do
    u:=%[x[i1]*x[i2]*Random(Fq) : i1,i2 in [1..t+s] | i1 le i2]+
        %[x[i1]*x[i2]*Random(Fq) : i1 in [1..t+s], i2 in [t+s+1..n]]+
        %[x[i1]*Random(Fq) : i1 in [1..n]]+Random(Fq);
    U:=Append(U,u);
end for;

// compute U(x-q(y),y)
Ug:=[];
for j in [1..m-t] do
    ug:= Evaluate(U[j],[x[i]-qq[i] : i in [1..t]] cat [x[i] : i in [t+1..n]]);
    Ug:=Append(Ug,ug);
end for;

// compute G
G:=[x[i]- qq[i] : i in [1..t]] cat Ug;

// compute A1 and A2 random affine linear bijections
L1:=Random(GLm); c1:=Random(Vm);
A1:=[%[L1[j][i]*z[i] : i in [1..m]]+c1[j]: j in [1..m]];
L2:=Random(GLn); c2:=Random(Vn);
A2:=[%[L2[j][i]*x[i] : i in [1..n]]+c2[j]: j in [1..n]];

// compute Gpub

```

```

GA2:=[(Evaluate(G[i],A2)) : i in [1..m]];
Gpub:=[Evaluate(A1[i],GA2) : i in [1..m]];

// target input and corresponding output to attack
secIPT:=Random(Vn);
OPT:=[Evaluate(Gpub[i],Eltseq(secIPT)) : i in [1..m]];

```

A.2 The Code for Method 1: Gröbner bases algorithms

The following code produces the output from §3.1. Theorem 1 implies that, if we compute the quadratic equations corresponding to $x - q(y)$ by Gaussian elimination, then reduce G_{pub} modulo a Gröbner basis of the space of quadrics that we found, this produces the desired quadratic system. We follow this approach in the toy example, even though it results in slightly more computations than necessary, as this allows us to take advantage of the pre-implemented functions in Magma [1].

```

Gpubeval:=[Gpub[i]-OPT[i] : i in [1..m]];

// recover the quadratic equations in Gpubeval by solving a linear system
Mon34:=[x[i1]*x[i2]*x[i3]*x[i4] :
        i1,i2,i3,i4 in [1..n] | i1 le i2 and i2 le i3 and i3 le i4]
        cat [x[i1]*x[i2]*x[i3] : i1,i2,i3 in [1..n] | i1 le i2 and i2 le i3];
Acf:=[];
for f in Gpubeval do
    cf:=[MonomialCoefficient(f,mon34) : mon34 in Mon34];
    Acf:=Append(Acf,cf);
end for;
AcfM:=Matrix(Acf);
sl,K:=Solution(AcfM,VectorSpace(Fq,#cf)!0);

// Gp2 contains the quadratic equations
// Gp4 contains the other equations (cubic and quartic)
Gp2:=[]; Gp4:=[];
B1:=Basis(K);
BB:=ExtendBasis(B1, Vm); B2:=[BB[i] : i in [#B1+1 .. m]];
for bb in B1 do
    Gp2:=Append(Gp2, &+[bb[j]*Gpubeval[j] : j in [1..m]]); end for;
for bb in B2 do
    Gp4:=Append(Gp4, &+[bb[j]*Gpubeval[j] : j in [1..m]]);
end for;

I2:=Ideal(Gp2);
GB2:=GroebnerBasis(I2);
BR:=[b: b in GB2 | TotalDegree(b) le 4];

```

```

// Ured contains the reduced equations (w.r.t. the quadratic equations)
Ured:=[];
for f in Gp4 do
    g:=NormalForm(f,GB2);
    Ured:=Append(Ured,g);
end for;

// Gred1 is the reduced quadratic system
Gred1:= Gp2 cat Ured;

```

A.3 The Code for Method 2: Higher Order Linearization Equations (HOLE)

```

tot:=(n+m+1)*(n+m+2) div 2; // lower bound on the number of evaluations to perform

// recover the quadratic relations between input and output variables
E:=[];
for count in [1..tot+20] do
    ipt:=Random(Vn);
    opt:=[Evaluate(Gpub[i],Eltseq(ipt)) : i in [1..m]];
    cf:=[opt[i1]*opt[i2] : i1,i2 in [1..m] | i1 le i2]
        cat [opt[i1]*ipt[i2] : i1 in [1..m], i2 in [1..n]]
        cat [ipt[i1]*ipt[i2] : i1,i2 in [1..n] | i1 le i2]
        cat opt cat Eltseq(ipt) cat [1];
    E:=Append(E,cf);
end for;

VV:=VectorSpace(Fq,#E);
EM:=Transpose(Matrix(E));
sol,K:=Solution(EM, VV!0);
BE:=Basis(K);

// given the target output, construct the equations for the target input
// Gred2 is the reduced quadratic system
Gred2:=[];
for bb in BE do
    cf:=[OPT[i1]*OPT[i2] : i1,i2 in [1..m] | i1 le i2]
        cat [OPT[i1]*x[i2] : i1 in [1..m], i2 in [1..n]]
        cat [x[i1]*x[i2] : i1,i2 in [1..n] | i1 le i2] cat OPT cat x cat [1];
    newf:="#+[bb[i]*cf[i] : i in [1..#cf]];
    Gred2:=Append(Gred2,newf);
end for;

```

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